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2 **Supporting Information for**

3 **Robust inference of causality in high-dimensional dynamical processes from the Information** 4 **Imbalance of distance ranks**

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Supporting Information Text

1. Information-theoretic formulation of the method

In the following we show that the Information Imbalance defined in Eq. (1) can be interpreted as the upper bound of an information-theoretic measure.

A. Copula variables. The bridge between the Information Imbalance measure and information theory rests on the statistical theory of copulas. Given a two-dimensional random variable $\mathbf{R} = (R_A, R_B)$ with joint density $p_{AB}(R_A, R_B)$ and marginal densities $p_A(R_A)$, $p_B(R_B)$, we define its *copula variable* as

$$\mathbf{C} = (c_A, c_B) = (F_A(R_A), F_B(R_B)) , \quad [1]$$

where F_A is the cumulative distribution function of R_A : $F_A(R) = \int_{-\infty}^R dR' p_A(R')$ (similarly for F_B). The relevance of copula variables lies in the two following results:

- the marginals of $\mathbf{C}$ are uniformly distributed in $[0, 1]$, result known as *probability integral transform* (1):

$$c_A \sim U_{[0,1]} , \quad c_B \sim U_{[0,1]} ; \quad [2]$$

- according to Sklar's theorem (2), the joint density of $\mathbf{R}$ can be decomposed as the product of the marginal densities and a joint copula distribution:

$$p_{AB}(R_A, R_B) = \tilde{p}_{AB}(c_A, c_B) p_A(R_A) p_B(R_B) . \quad [3]$$

In particular, Sklar's theorem implies that the joint entropy of (R_A, R_B) can be decomposed as the sum of the marginal entropies and the joint entropy of the copula variables (c_A, c_B) :

$$H(R_A, R_B) = H(R_A) + H(R_B) + H(c_A, c_B) . \quad [4]$$

From Eq. (2) it follows that the marginal entropy of each copula variable is zero; as a consequence,

$$H(c_A, c_B) = H(c_B | c_A) = H(c_A | c_B) . \quad [5]$$

Given these properties, the mutual information between variables R_A and R_B can be expressed equivalently as the negative conditional entropy of their copula variables (3, 4):

$$I(R_A, R_B) = H(R_A) + H(R_B) - H(R_A, R_B) \quad [6a]$$

$$= -H(c_A | c_B) = -H(c_B | c_A) \quad [6b]$$

B. The Information Imbalance as an upper bound of a mutual information. Given a data set $\{x_i\}_{i=1}^N$ drawn from the unknown probability density $p(x)$, from hereon we call R_A and R_B the RVs describing pair-wise distances measured with the metrics d_A and d_B , respectively, and we denote with R_{ij}^A and R_{ij}^B their empirical realizations:

$$R_{ij}^A \doteq d_A(x_i, x_j) , \quad [7a]$$

$$R_{ij}^B \doteq d_B(x_i, x_j) . \quad [7b]$$

With a change of variable, the probability density of the distances from a point x_i in space d_A can be written as

$$p(R_A | x_i) \doteq p(R_A^i) = \int d\tilde{x} p(\tilde{x}) \delta(d_A(\tilde{x}, x_i) - R_A^i) , \quad [8]$$

where the integral is over the support of $p(x)$ and δ denotes the Dirac delta function. The distribution $p(R_B^i)$ can be defined analogously. Given these distributions, the relative information content of the two distance spaces conditioned on x_i can be measured by the mutual information $I(R_A^i, R_B^i)$, which can be cast into a conditional copula entropy according to Eqs. (6b). However, if we aim at quantifying how well each distance can predict the other, the mutual information is not a suitable measure. Indeed, we expect that the descriptive power of distance d_A with respect to distance d_B will change if we invert the roles of the two spaces, while the mutual information is symmetric under the exchange of its arguments.

Guided by the intuition that a “good” distance should first and foremost describe points that are “similar” as close, in ref. (5) we proposed to introduce an asymmetry by defining a *restricted mutual information*:

$$I^\varepsilon(R_A^i \rightarrow R_B^i) \doteq - \int_0^\varepsilon d\tilde{c}_A^i H(c_B^i | c_A^i = \tilde{c}_A^i) . \quad [9]$$

Here c_A^i is the copula variable associated to R_A^i and its empirical values can be simply computed as $\{\frac{1}{N} r_{ij}^A\}$, where r_{ij}^A is the rank of point j relative to point i according to distance d_A (similarly in space B). This reflects the fact that distance ranks and copulas are statistically identical variables. In the limit of small ε , the restricted mutual information of Eq. (9) quantifies the information shared by the distances d_A and d_B , constraining the first to a small neighborhood around x_i . With the assumption

that large distances are noninformative, we interpret this measure as the actual information in space d_B which is contained in d_A . If we expand the restricted mutual information in the limit $\varepsilon \rightarrow 0$, the first nonzero term is its first derivative around zero:

$$\lim_{\varepsilon \rightarrow 0} \frac{I^\varepsilon(R_A^i \rightarrow R_B^i)}{\varepsilon} = \frac{\partial}{\partial \varepsilon} I^\varepsilon(R_A^i \rightarrow R_B^i)|_{\varepsilon=0} \quad [10a]$$

$$= - \lim_{\varepsilon \rightarrow 0} H(c_B^i | c_A^i = \varepsilon) \quad [10b]$$

$$= \lim_{\varepsilon \rightarrow 0} \int_0^1 dc_B^i p(c_B^i | c_A^i = \varepsilon) \log p(c_B^i | c_A^i = \varepsilon). \quad [10c]$$

In our finite data set, we can translate the condition $c_A^i = \varepsilon$ in the limit of small ε by taking the first k neighbors of i , i.e. the points j such that $c_{ij}^A \leq \frac{k}{N}$. Since the number k should be much smaller than N to fulfill the continuous condition $\varepsilon \rightarrow 0$, we can only access a small number of samples of the density $p(c_B^i | c_A^i = \varepsilon)$, and a direct estimate of Eq. (10b) would be unreliable. Instead of trying to compute Eq. (10b) explicitly, we suppose to know only the average of $p(c_B^i | c_A^i = \varepsilon)$, that we estimate in the limit $\varepsilon \rightarrow 0$ as

$$\langle c_B^i | c_A^i = \varepsilon \rangle \simeq \frac{1}{N} \langle r_B^i | r_A^i = \varepsilon \rangle := \frac{1}{Nk} \sum_{\substack{j \\ r_{ij}^A \leq k}} r_{ij}^B, \quad [11]$$

and we look for an upper bound of the copula entropy in Eq. (10b), using a maximum entropy approach. We first observe that the average in Eq. (11) tends to 0 in the case of d_A being maximally informative with respect to d_B , and to 1/2 in the fully non-informative case (namely when $p(c_B^i | c_A^i = \varepsilon) = p(c_B^i)$). The Information Imbalance defined in Eq. (1) in the main text is then a rank-based estimate of this quantity, with an additional global average over all the data points:

$$\Delta(A \rightarrow B) \simeq 2 \left\langle \lim_{\varepsilon \rightarrow 0} \langle c_B^i | c_A^i = \varepsilon \rangle \right\rangle_{i=1, \dots, N}. \quad [12]$$

It is well known that among all the probability densities with support $[0, +\infty)$ and assigned mean, the one with maximum entropy is exponential (6). The same can be shown in the case of distributions with support $[0, 1]$, by using the technique of Lagrange multipliers. In particular, the probability distribution maximizing the entropy of Eq. (10b) is

$$p_{ME}(c_B^i | c_A^i = \varepsilon) = \frac{\lambda}{1 - e^{-\lambda}} e^{-\lambda c_B^i}, \quad [13]$$

where the parameter λ is related to the mean by

$$\langle c_B^i | c_A^i = \varepsilon \rangle = \frac{1}{\lambda} + \frac{1}{1 - e^{-\lambda}}. \quad [14]$$

By substitution we find a first upper bound:

$$\lim_{\varepsilon \rightarrow 0} H(c_B^i | c_A^i = \varepsilon) \leq -\log \frac{\lambda}{1 - e^{-\lambda}} + 1 + \frac{\lambda}{1 - e^{-\lambda}}. \quad [15]$$

The right-hand side of Eq. (15) is further maximized by a quantity directly related to the average in Eq. (14):

$$\lim_{\varepsilon \rightarrow 0} H(c_B^i | c_A^i = \varepsilon) < \log \left(\lim_{\varepsilon \rightarrow 0} \langle c_B^i | c_A^i = \varepsilon \rangle \right) + 1. \quad [16]$$

In particular, the right-hand sides of Eqs. (15) and (16) are asymptotically equal in the limit $\lambda \rightarrow 0$, namely when $\langle c_B^i | c_A^i = \varepsilon \rangle \rightarrow 0$. This is the case of d_A being maximally informative with respect to d_B . The last inequality, aside with Eq. (12), allows to write the Information Imbalance as an upper bound of an information-theoretic measure:

$$\Delta(A \rightarrow B) \gtrsim 2 \left\langle \exp \left(\lim_{\varepsilon \rightarrow 0} H(c_B^i | c_A^i = \varepsilon) - 1 \right) \right\rangle_{i=1, \dots, N}. \quad [17]$$

Equivalently, in terms of the restricted mutual information, the inequality takes the form

$$\Delta(A \rightarrow B) \gtrsim 2 \left\langle \exp \left(- \lim_{\varepsilon \rightarrow 0} \frac{I^\varepsilon(R_A^i \rightarrow R_B^i)}{\varepsilon} - 1 \right) \right\rangle_{i=1, \dots, N}. \quad [18]$$

Due to the asymptotic relationship mentioned above, the Information Imbalance provides a better estimate of the right-hand side of Eq. (17) in the informative regime, namely when $\Delta(A \rightarrow B)$ is close to 0.

C. Interpretation of the upper bound. A straightforward interpretation of the restricted mutual information introduced in Eq. (9) in the limit $\varepsilon \rightarrow 0$ can be provided by rewriting Eq. (10b) in terms of the distance variables R_A^i and R_B^i . Applying Sklar's theorem and defining $\eta = F_A^{-1}(\varepsilon)$ we can write

$$\lim_{\varepsilon \rightarrow 0} \frac{I^\varepsilon(R_A^i \rightarrow R_B^i)}{\varepsilon} = \lim_{\eta \rightarrow 0} \int_0^{+\infty} dR_B^i p(R_B^i | R_A^i = \eta) \log \frac{p(R_B^i | R_A^i = \eta)}{p(R_B^i)} \quad [19a]$$

$$= \lim_{\eta \rightarrow 0} D_{KL}[p(R_B^i | R_A^i = \eta) \| p(R_B^i)], \quad [19b]$$

where D_{KL} denotes the Kullback-Leibler divergence, which is a pseudodistance between probability densities. Eq. (19b) clarifies the concept of shared information among distance spaces: when we condition the distribution of distances in d_B to close points according to d_A , the greater the change in the distribution, the more informative d_A is with respect to d_B . This allows to translate in an information-theoretic framework the simple intuition that d_A carries information about d_B when close points in the former remain close in the latter.

If we specialize to the problem of causal detection by taking the distance spaces $d_A = d_{\alpha X(0), Y(0)}$ and $d_B = d_{Y(\tau)}$, the optimization over α of the right-hand side of Eq. (17) takes the form of a *minimum entropy* protocol, where we aim to maximize the predictability of the future distances in the space of the effect variable, when we consider points that were close in the past according to both the effect and the causal variables.

2. Other approaches for causality detection

A. Measure L . In (7), Chicharro and Andrzejak introduced a rank-based measure to detect unidirectional couplings in non-synchronizing conditions, lying on the idea of state space reconstruction using time-delay embeddings (in the following, $\tilde{x}(t)$ and $\tilde{y}(t)$). To write the expression of this statistical measure, we call v_{tj} and w_{tj} the time indices of the j -th nearest neighbors of $\tilde{x}(t)$ and $\tilde{y}(t)$, respectively, from which we exclude temporal neighbors within a window of size W (requiring $|v_{tj} - t| > W$ and $|w_{tj} - t| > W$). Distances are measured in the shadow manifolds according to Euclidean metrics, denoted by d_X and d_Y . Considering k nearest neighbors for each point, the *measure L* is defined as

$$L(X | Y) \doteq \frac{1}{N} \sum_{t=1}^N \frac{G_t(X) - G_t^k(X | Y)}{G_t(X) - G_t^k(X)}, \quad [20]$$

where $G_t(X)$, $G_t^k(X)$ and $G_t^k(X | Y)$ are respectively the mean rank, the minimal mean rank and the Y -conditioned mean rank: $G_t(X) = \frac{N}{2}$, $G_t^k(X) = \frac{k+1}{2}$ and $G_t^k(X | Y) = \frac{1}{k} \sum_{j=1}^k r_{t, w_{tj}}^X$ ($r_{tt'}^X$ denotes the distance rank of $\tilde{x}(t')$ with respect to $\tilde{x}(t)$). Taking only the first nearest neighbor ($k = 1$), it is easy to show that the measure L is related to the standard Information Imbalance by a straightforward linear relation:

$$L(X | Y) = \frac{1}{N} \sum_{t=1}^N \frac{N/2 - G_t^1(X | Y)}{N/2 - 1} \quad [21a]$$

$$= \frac{N}{N-2} - \frac{2}{N-2} \frac{1}{N} \sum_{t=1}^N G_t^1(X | Y) \quad [21b]$$

$$= \frac{N}{N-2} - \frac{2}{N-2} \langle r_X | r_Y = 1 \rangle \quad [21c]$$

$$= \frac{N}{N-2} [1 - \Delta(d_Y \rightarrow d_X)]. \quad [21d]$$

Using a generic number of neighbors k , this relation becomes

$$L(X | Y) = \frac{N}{N-k-1} [1 - \Delta(d_Y \rightarrow d_X)]. \quad [22]$$

We notice that the Information Imbalance of Eq. (21d) is referred to distance spaces with no relative time lag ($\tau = 0$), which is a major difference between state space reconstruction methods such as the measure L and methods based on predictability. In (7) the authors proposed to assess the presence of the coupling $X \rightarrow Y$ by checking whether the condition $\Delta L = L(X | Y) - L(Y | X) > 0$ is verified. In terms of the Information Imbalance measure, this translates into the inequality $\Delta(d_Y \rightarrow d_X) < \Delta(d_X \rightarrow d_Y)$.

For the analysis reported in Fig. 3 in the main text we chose the optimal values of the embedding time τ_e following the mutual information criterion, except for the case of the Lorenz 96 systems, where we found $\tau_e = 1$ to minimize the difference between the signals in the $X \rightarrow Y$ and $Y \rightarrow X$ couplings. We set $k = 5$ both for the Rössler and the Lorenz systems, and $k = 20$ for the Lorenz 96 systems.

B. Extended Granger causality. Given two one-dimensional signals $x(t)$ and $y(t)$, the Extended Granger Causality method (8) tries to assess whether x has a causal influence on y by studying which one among the two autoregressive models

$$y(t + \tau_e) = \sum_{j=1}^E \alpha_j y(t + (j-1)\tau_e) + \varepsilon_y, \quad [23a]$$

$$y(t + \tau_e) = \sum_{j=1}^E \left[\beta_j y(t + (j-1)\tau_e) + \gamma_j x(t + (j-1)\tau_e) \right] + \varepsilon_{y|x}, \quad [23b]$$

gives the best description of the unknown dynamics, performing several local regressions according to Eqs. (23) instead of a single global fit as in standard Granger causality. Following ref. (9), in all our analysis we carried out 200 local regressions. Each local regression is carried out by sampling a random point in the space of time-delay embeddings and considering the neighbor points within a radius δ from the central one, as measured by the Euclidean distance in the space of the joint embeddings $(x(t), \dots, x(t - (E-1)\tau_e), y(t), \dots, y(t - (E-1)\tau_e))$. Equivalently, in this work we defined the neighborhood size by fixing the number of neighbors k . The residuals estimated in each local regression are used to compute the Extended Granger Causality index $\Delta_{x \rightarrow y} = \langle 1 - \varepsilon_{y|x} / \varepsilon_x \rangle$, which identifies the presence of a coupling when statistically different from zero.

In the tests reported in Fig. 3 in the main text we chose k using a single experiment at fixed ε for each pair of systems, and then we used it to compute the extended Granger causality index for all the experiments. We repeated these steps for several choices of the time-delay embedding parameters, selecting $k = 200$, $E = 3$ and $\tau_e = 1$ both for the Rössler and the Lorenz systems, and $k = 500$, $E = 30$ and $\tau_e = 7$ for the Lorenz 96 systems.

As the method was proposed for time series only, we adapted it to the case where x is a static variable by replacing Eq. (23b) with

$$y(t + \tau_e) = \sum_{j=1}^E \beta_j y(t + (j-1)\tau_e) + \gamma x + \varepsilon_{y|x} \quad [24]$$

in order to apply it when x is the duration of the comparison stimulus in the EEG experiment. Since the single EEG time series do not satisfy the requirement of stationarity, similarly to our approach we constructed a data set for each t and each participant using the different trials as independent realizations.

While in our method τ and τ_e are distinct parameters, the Extended Granger Causality approach is constructed with $\tau = \tau_e$. For this reason, instead of studying the behavior of the Extended Granger Causality index as a function of the separation τ between the predictive window and the predicted point, we fixed τ_e and scanned the time t of the predicted point, moved concurrently with the predictive window. In the analysis carried out with our approach, by contrast, the predictive window was kept fixed after the stimulus onset / offset.

In the EEG analysis we found the best results for $E = 3$ and setting k to the total number of points available for each subject, namely performing a single global regression (insets of Fig. 4A and B in the main text). Notably, with these parameters the Extended Granger Causality is equivalent to the standard Granger causality test.

C. Convergent Cross Mapping. The method was implemented as described in ref. (10), using neighborhoods of size $E + 1$ in the shadow manifolds of x_1 and y_1 to compute the cross mapping coefficients. The cross-map skill, quantified by the Pearson correlation ρ between the reconstructed and the ground-truth points in spaces x_1 and y_1 , was studied as a function of the length L of the trajectory in order to select ρ after convergence. In particular, we observed convergence at $L = 30000$ for the identical Rössler systems ($E = 3$, $\tau_e = 5$), $L = 20000$ for the different Rössler systems ($E = 3$, $\tau_e = 5$), $L = 40000$ for the bidirectionally coupled Lorenz systems ($E = 3$, $\tau_e = 3$) and $L = 50000$ for the Lorenz 96 systems ($E = 30$, $\tau_e = 7$). The values of τ_e were chosen as the first minima of the lagged mutual information (11).

D. Transfer Entropy. We employed the formulation provided by Paluš and Vejmelka in ref. (12), which presents the measure in the $X \rightarrow Y$ direction through the conditional mutual information $I(\tilde{x}(0); \tilde{y}(\tau) | \tilde{y}(0))$. We implemented this measure with the conditional mutual information estimator proposed in ref. (13) and implemented in the Python library knncmi, setting $k = 3$ for all the tested systems and using the following time-delay embedding parameters: $E = 3$, $\tau_e = 1$ for the Rössler and Lorenz systems, and $E = 30$, $\tau_e = 1$ for the Lorenz 96 systems. The time lag was set to $\tau = 20$ for the Rössler systems, $\tau = 5$ for the Lorenz systems and $\tau = 30$ for the Lorenz 96 systems.

3. Generalization of the Imbalance Gain to three systems

The Imbalance Gain in direction $X \rightarrow Y$ defined in Eq. (3) of the main text can be generalized in order to take into account the effects induced by a third system Z :

$$\delta\Delta(X \rightarrow Y | Z) \doteq 1 - \frac{\min_{\alpha_X, \alpha_Z} \Delta(d_{\alpha_X X(0), \alpha_Z Z(0), Y(0)} \rightarrow d_{Y(\tau)})}{\min_{\alpha_Z} \Delta(d_{\alpha_Z Z(0), Y(0)} \rightarrow d_{Y(\tau)})}. \quad [25]$$

We call this measure *conditional* Imbalance Gain, as it quantifies how much adding the information of X at time $t = 0$ can improve the predictability of Y at time $t = \tau$, given a baseline prediction which already includes the third system Z at time

$t = 0$. Similarly to the two-system case, we claim that X causes Y given Z when $\delta\Delta(X \rightarrow Y | Z) > 0$. The scaling parameter α_X plays the same role of α in Eq. (3), while α_Z is an additional optimization parameter which scales the variables of system Z . The optimal self-prediction in the unconditioned case, $\Delta(\alpha = 0)$, is here replaced by $\min_{\alpha_Z} \Delta(d_{\alpha_Z Z(0), Y(0)} \rightarrow d_{Y(\tau)})$. If the Imbalance Gain of Eq. (3) can be seen as a generalization of Granger causality, the conditional Imbalance Gain corresponds to its multivariate formulations (see for example refs. (8, 14)).

We tested the conditional measure on three different Rössler systems, with parameters $\omega_X = 1.015$, $\omega_Y = 0.985$ and $\omega_Z = 1.005$, considering the following coupling scenarios (one for each row of Fig. S1):

1. X and Y are unidirectionally coupled and Z interacts neither with X nor with Y ;
2. X and Y are indirectly coupled through Z ;
3. X and Y are neither directly nor indirectly coupled and Z is a common driver.

The results reported in Fig. S1 were obtained using all the coordinates of the three systems and averaging both the unconditioned and the conditional measures over 20 independent estimates. Similarly to the unconditioned case (see Materials and Methods - Average Imbalance Gain), in the conditional measure we averaged the α_X -dependent Imbalance Gains obtained by optimizing only the α_Z parameter,

$$\delta\Delta(X \rightarrow Y | Z)(\alpha_X) \doteq 1 - \frac{\min_{\alpha_Z} \Delta(d_{\alpha_X X(0), \alpha_Z Z(0), Y(0)} \rightarrow d_{Y(\tau)})}{\min_{\alpha_Z} \Delta(d_{\alpha_Z Z(0), Y(0)} \rightarrow d_{Y(\tau)})}, \quad [26]$$

inferring a single optimal α_X for all the independent estimates.

While in scenario 1) the conditional Imbalance Gain $\delta\Delta(X \rightarrow Y | Z)$ coincides with the standard measure $\delta\Delta(X \rightarrow Y)$ (panels A and B), in scenarios 2) and 3) it allows to clearly understand that X and Y are not directly coupled. This shows that our approach allows to exclude unambiguously the presence of a direct causal link from X to Y both when a common driver is present and when the link is indirect, opening up to applications in causal network reconstruction for time-dependent variables (14).

4. Stability with respect to noise of the Imbalance Gain

Real-world time series can be “noisy” as a result of measurement errors or as a consequence of a stochastic underlying dynamics. We investigated these two scenarios separately, both adding noise *a posteriori* to the deterministic trajectories (Fig. S2A) and by adding it at each integration step (Fig. S2B). In the latter case we only considered the case of Gaussian and uncorrelated (white) noise, which turns the dynamical equations into Langevin-like equations. The use of correlated noise may hinder the performance of the method and is expected to affect the behavior of the Imbalance Gain as a function of τ , depending on the characteristic time-scales of the noise. To avoid mixing the effect of noise with other nuisance sources, we computed the Imbalance Gain considering all the coordinates of the two systems.

As expected, in both scenarios the signal in the $X \rightarrow Y$ direction appears to weaken as the noise level increases. In direction $Y \rightarrow X$, where the coupling is absent, the signal is negligible up to a 15% noise level in the first case (Fig. S2A), while the Imbalance Gain remains close to zero for all the tested noise amplitudes in the second scenario. We interpret this difference as follows. When the noise is added *a posteriori* on a deterministic signal, its effect is to reduce the information content of the distance space $d_{X(0)}$ with respect to $d_{X(\tau)}$, so that the former is not anymore the most informative metric that one can build with respect to the latter by using the dynamic variables at time $t = 0$. In this scenario the self-prediction of X becomes non-optimal, leaving the possibility of improvements by including the information on $Y(0)$ in the distance space $d_{\alpha_Y(0), X(0)}$. As Fig. S2A shows, the reduction of information induced by measurement noise in $d_{X(0)}$ results in a nonzero Imbalance Gain in direction $Y \rightarrow X$. When the underlying dynamics is stochastic, instead, the effect of noise is to reduce the predictability of $d_{X(\tau)}$ given the knowledge of $d_{X(0)}$, because the dynamics of X is altered and the system exhibits a shortened memory. In this context, the distance space $d_{X(0)}$ still remains the most informative that one can construct to make a prediction about $d_{X(\tau)}$ with the dynamic variables at time $t = 0$. As a consequence, the information about $Y(0)$ keeps being redundant as in the case of a deterministic dynamics, and in Fig. S2B we observe consistently that the signal in direction $Y \rightarrow X$ is only slightly affected by the amplitude of the integration noise.

5. Role of the optimization parameter

In the presentation of the method we suggested that the scaling parameter α in the distance $d_{\alpha X(0), Y(0)}$ is necessary to account for the magnitude of the coupling strength in direction $X \rightarrow Y$. In Fig. S3 we provide a numerical evidence of this claim by showing that for the dynamical systems not affected by synchronization (namely the bidirectionally coupled Lorenz systems and the unidirectionally coupled Lorenz 96 systems of Fig. 3) the optimal value of α in the minimum is a monotonically increasing function of the coupling strength, with small fluctuations likely due to statistical noise. This property allows to use the optimal values of α for comparing different coupling strengths in two scenarios involving the same dynamical systems, but it does not imply that the absolute coupling strength can be inferred from the knowledge of the optimal α . As a simple counterexample, an arbitrary re-definition of the units of X by a scaling factor 2 ($X \mapsto 2X$) would scale the optimal α by a factor $1/2$ ($\alpha \mapsto 1/2 \alpha$), even if the coupling strength in direction $X \rightarrow Y$ is untouched. The optimal scaling parameters displayed in Fig. S3 were obtained by using 20 independent realizations of the same dynamics, as described in the Materials and Methods - Average Imbalance Gain.

6. Assessing the statistical significance of the Imbalance Gain for a single estimate

Due to finite-sample effects, the Imbalance Gain (Eq. (3) in the main text) can be small but nonzero even in absence of causality. This requires a quantitative approach to evaluate whether a single estimate of the Imbalance Gain is compatible with zero.

The statistical significance of our measure can be assessed by a permutation test which generates a null distribution under the null hypothesis of “absence of causality”. If we consider the direction $X \rightarrow Y$, we can generate the null distribution of the Imbalance Gain $\delta\Delta(X \rightarrow Y)$ under the hypothesis H_0 “ X does not cause Y ” by applying random permutations to the data point indices (i.e. the trajectories in the ensemble) in the only space X . For example, if trajectory i includes the components $X^i = (x_1^i, x_2^i, x_3^i)$ and $Y^i = (y_1^i, y_2^i, y_3^i)$, we denote by σ a permutation that only affects the trajectory index in the X space: $X^{\sigma(i)} = (x_1^{\sigma(i)}, x_2^{\sigma(i)}, x_3^{\sigma(i)})$. The Imbalance Gain in direction $X \rightarrow Y$ under such a random permutation,

$$\delta\Delta^\sigma(X \rightarrow Y) \doteq \frac{\Delta(d_{Y(0)} \rightarrow d_{Y(\tau)}) - \min_\alpha \Delta(d_{\alpha X(0)^\sigma, Y(0)} \rightarrow d_{Y(\tau)})}{\Delta(d_{Y(0)} \rightarrow d_{Y(\tau)})}, \quad [27]$$

is by construction compatible with zero independently of the ground-truth coupling scenario. Indeed, permuting the indices results in a randomization of the distances in space $d_{X(0)}$, eliminating any information that this space could provide about distances measured in $d_{Y(\tau)}$. Therefore, computing Eq. (27) under many random permutations in space $X(0)$ allows sampling the null distribution $p_{H_0}(\delta\Delta)$, from which we can compute the p -value of the actual Imbalance Gain and accept (reject) the null hypothesis when p is larger (smaller) than a chosen confidence. The same approach can be applied in direction $Y \rightarrow X$ by permuting the point indices within the Y space only.

We measured the statistical significance of single Imbalance Gain estimates for the unidirectionally coupled Lorenz 96 systems (the same employed in the comparisons of Fig. 3 in the main text). As in the main text, we considered 31 values of the coupling strength ε equally spaced in the interval $[0, 1.5]$. Tables S1A and B report the confusion matrices associated to the outcome of the statistical test, both using all the 40 coordinates of the systems and with time-delay embeddings ($E = 30$). The rate of false positive detections is below 10% in both cases. Using time-delay embeddings the rate of false negative detections increases to 16.7%, but all the wrong detections are limited to the weak coupling regime ($\varepsilon < 0.25$).

7. Assessing the false positive rate of repeated and independent estimates

For each measure used in the comparison of Fig. 3, the false positive rate was assessed over systems coupled in direction $X \rightarrow Y$ by testing the direction without causal link ($Y \rightarrow X$) according to the following steps:

1. The measure in direction $Y \rightarrow X$ was repeatedly computed over 20 independent realizations of the dynamics, thus providing 20 independent estimates $\{m_i\}$ ($i = 1, \dots, 20$). The independent realizations were obtained by integrating the dynamical equations from different initial conditions and discarding the first 100000 points of each integration.
2. For each value of ε , the quantity $t = \langle m \rangle / \text{SE}[\langle m \rangle]$ was computed, where $\text{SE}[\langle m \rangle] = \text{Std}[m] / \sqrt{20}$ denotes the standard error of the mean. This statistic asymptotically follows a Student’s t -distribution with $20 - 1$ degrees of freedom.
3. In order to assess if $\langle m \rangle$ is compatible with zero, a Student’s t -test employing the estimate of t in the previous point was carried out. According to this test, the null hypothesis “the population mean of the causal measure m is zero in direction $Y \rightarrow X$ ” is rejected when $t > \text{threshold}$. We selected the threshold value in order to reject the null hypothesis with p -value $p < 0.001$, but we also studied the sensitivity of the test as a function of p , as reported in Fig. S4. Since $\langle m \rangle$ is non-negative for all the causality measures, we employed a one-tailed test where p corresponds to the probability of sampling a value larger than t , namely we only considered the right tail of the distribution.

After performing the t -test for each value of ε , the false positive rate was computed as the fraction of ε values resulting in a rejection of the null hypothesis.

8. EEG data

A. Experimental details. Volunteers underwent six experimental blocks, each composed by 60 trials. In each trial, a visual stimulus, varying in display time trial by trial, was presented at the center of the screen (Samsung 22” SyncMaster 2233RZ, 1680x1050 pixels resolution, 120 Hz refresh rate, viewing distance 60 cm, grey background color). This stimulus, called comparison, was a sinusoidal annulus grating (5 degrees of visual angle outer-circle diameter, 1 degree of visual angle inner-circle diameter, 1 cycle per degree, 100% contrast) composed by 5 concentric parts of equal area slowly and rigidly rotating around the center of the screen (~ 0.6 rad/s with alternating rotation senses, either clockwise or counterclockwise). The comparison stimulus could be displayed for 0.3, 0.4, 0.5, 0.6, 0.7, 0.8 or 0.9 seconds in each trial and it was followed (after an interval uniformly distributed between 0.5 and 1 second) by an auditory stimulus of 0.6 seconds (burst of white noise delivered through Etymotic Research ER1 Insert Earphones). Each stimulus duration was tested 10 times in each block. Volunteers were instructed to maintain their gaze on a fixation cross (0.2 degrees of visual angle in diameter) presented at the center of the screen throughout the experiment and to report, by pressing a button on the keyboard, whether the first or the second stimulus was presented for longer time. After providing the response the next trial started automatically after an interval uniformly

distributed between 0.5 and 1 second. Volunteers received no feedback on their performance. Volunteers performed the task comfortably sitting in front of the apparatus while we recorded their EEG activity using 64 active electrodes (arranged in a 10-20 layout, GND in Fpz and REF in FCz, cap model: actiCAP slim) BrainProducts actiCHamp Plus (Brain Products GmbH, Gilching, Germany) with 2048 Hz sampling frequency. No filtering was applied during the recording. In order to minimize head motion and blink artifacts, volunteers placed their head on a chin rest and they were instructed to blink when providing the response.

B. Data preprocessing. EEG data preprocessing was performed using EEGLAB toolbox (15) in Matlab (Mathworks Inc.). First, we removed motion and blink artifacts using automatic labeling of independent components (IC) using the ICLabel toolbox (16). Only ICs with a probability higher than 90% of being correctly classified as muscle or eye artifact were removed. Then, the continuous EEG signal was downsampled to 250 Hz sampling rate, referenced to a common average reference and filtered using a 4th-order Butterworth band-pass filter with range 0.1-40 Hz. The EEG data corresponding to the comparison stimulus was epoched starting from 200 ms before its onset to 500 ms after its offset. These epochs were detrended, and were examined to remove any remaining artifacts. We automatically removed epochs based on signal amplitude threshold, removing all trials in which the EEG amplitude exceeded $\pm 50 \mu V$. All the remaining trials were visually inspected to detect and remove the ones containing any residual artifacts. A total of $\sim 0.1\%$ of trials were discarded. Additionally, we excluded from further analyses temporal channels (T7, T8, TP7, TP8, TP9, TP10, P7, P8, FT7, FT8, FT9, FT10). We constructed then two data sets corresponding to the onset and offset period of comparison presentation, available at the link reported in the main text (Materials and Methods section - Data, Materials and software availability). The onset data set contained EEG traces of 300 ms length starting from stimulus onset baseline corrected using 44 ms pre-stimulus period. The offset data set consisted of EEG traces of 500 ms length starting from stimulus offset baseline corrected using the 88 ms window centered at stimulus offset (44 ms pre-offset and 44 ms post-offset period).

C. Validation of the Imbalance Gain measure with a dichotomous causal variable. As an additional test of our measure, we carried out a causal analysis where X is a dichotomous variable discriminating between trials where the stimulus is presented after a fixed time interval ($\Delta t = 80$ ms) and trials where the visual stimulus is not presented. Since the experiment did not include control trials where the visual stimulus was not presented to participants, the data set for this analysis was built from the original EEG time series with the following steps:

- for each participant half of the trials (labelled by $X = 1$) were shifted by 120 ms, setting the initial time $t = 0$ to -80 ms from the onset of the stimulus, while for the remaining trials (labelled by $X = 0$) the initial time was set to -200 ms from stimulus onset;
- a baseline correction was carried out using the initial 44 ms of the new series, namely using the EEG signal in range $[-200, -256]$ ms from stimulus onset for the trials of class $X = 0$, and the range $[-80, -36]$ ms for the series labelled by $X = 1$.

By employing time-delay embeddings of 44 ms, a trivial causality between X and the EEG signal should arise only for $\tau > 80 - 44$ ms, namely when the predicted windows cross the time point at which the trials of class $X = 0$ and $X = 1$ start to differentiate. Consistently, the Imbalance Gain shown in Fig. S6 becomes nonzero after the aforementioned time lag. The analysis was restricted to one of the occipital electrodes (POz), where the first cortical signature of a visual input should be detected (17).

9. Coupled dynamical systems with different time scales

An interesting dynamical scenario occurs when the driver system is particularly faster or slower than the driven system. In order to study how the Imbalance Gain and the other causality measures behave in such a scenario, we extended the equations of the unidirectionally coupled Lorenz 96 systems with an additional parameter $\tilde{\omega}$, which tunes the velocity of X :

$$\dot{X} = \tilde{\omega} f(X), \quad [28a]$$

$$\dot{Y} = g(Y) + G(X, Y). \quad [28b]$$

The explicit expressions of the functions f , g and G are reported in the Materials and Methods - Details on the dynamical systems. The systems tested in the last column of Fig. 3 correspond to the specific case $\tilde{\omega} = 1$. Increasing $\tilde{\omega}$ is equivalent to increase the velocity of X while keeping the velocity of Y untouched.

In Fig. S7 we report the behaviors of the different causality measures as a function of the parameter $\tilde{\omega}$. For all the methods we employed the same parameters used for the Lorenz 96 systems of Fig. 3, and we computed errors over 20 independent estimates. The critical observation, which makes the interpretation of the results nontrivial, is that the variation of $\tilde{\omega}$ appears to drastically affect the properties of system Y , changing its Lyapunov exponents and its chaoticity. Therefore, the effects of varying the relative velocity of X and Y on the causality measures are highly non-linear, as small changes of $\tilde{\omega}$ can result in driven systems Y with noticeably different dynamical properties.

Remarkably, also in this test we observe that all the causality measures, except the Imbalance Gain, are not able to recognize that causality in the $Y \rightarrow X$ direction is absent, bringing to false-positive rates of almost 100% (the false positives for the Imbalance Gain are at most 15% if one uses time-delay embeddings, panel B).

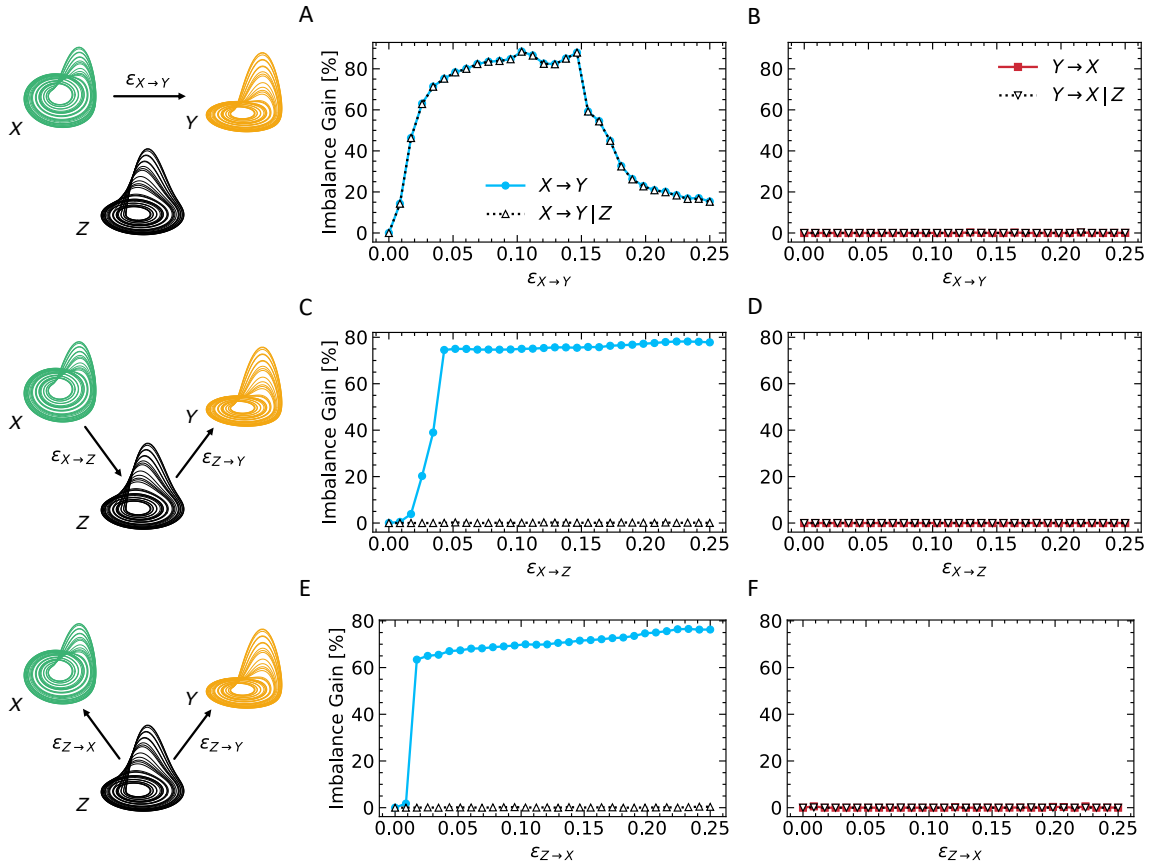


Fig. S1. Comparison between unconditional and conditional Imbalance Gains, for three different Rössler systems X , Y and Z . (A),(B): Unidirectional coupling $X \rightarrow Y$, Z uncoupled. (C),(D): Unidirectional couplings $X \rightarrow Z$ and $Z \rightarrow Y$, with $\varepsilon_{Z \rightarrow Y} = 0.05$. (E),(F): Unidirectional couplings $Z \rightarrow X$ and $Z \rightarrow Y$, with $\varepsilon_{Z \rightarrow Y} = 0.05$. Each point was obtained by averaging 20 different independent estimates of the Imbalance Gain. The minimization over α in the unconditioned measure was carried out using a grid of 50 evenly spaced points in $[0, 0.25]$, while the minimization over α_X and α_Z in the conditional Imbalance Gain was performed using a two-dimensional grid with 50×50 points in $[0, 0.25] \times [0, 0.25]$.

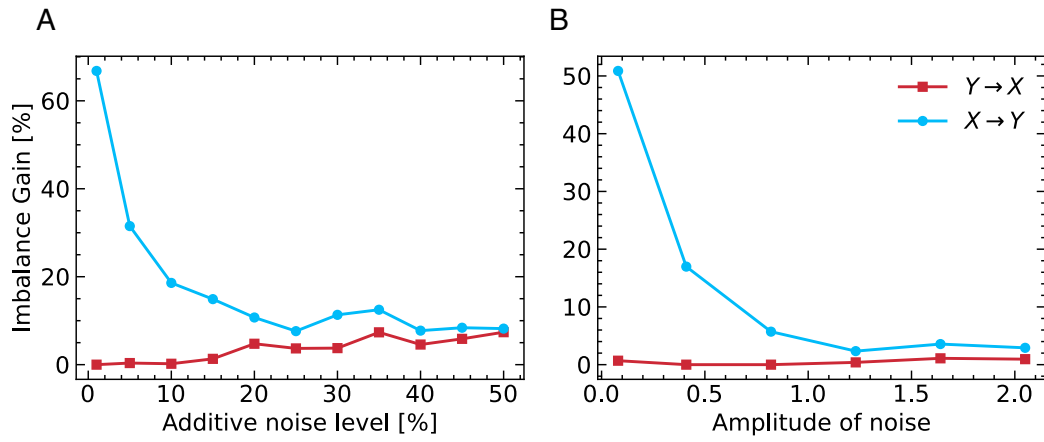


Fig. S2. Effect of noise on the Imbalance Gain, measured on the unidirectionally coupled Rössler systems ($X \rightarrow Y$) with coupling $\varepsilon = 0.1035$. (A) White noise was added a posteriori and independently to each dynamical variable, choosing its amplitude as a fraction (along the x -axis) of the standard deviation of the entire trajectory in the corresponding direction. (B) Independent white noises were added during the integration of the dynamical systems, only to the variables x_1 and y_1 , with amplitude reported along the x -axis. Both the tests were carried out with $k = 1$ and $\tau = 5$.

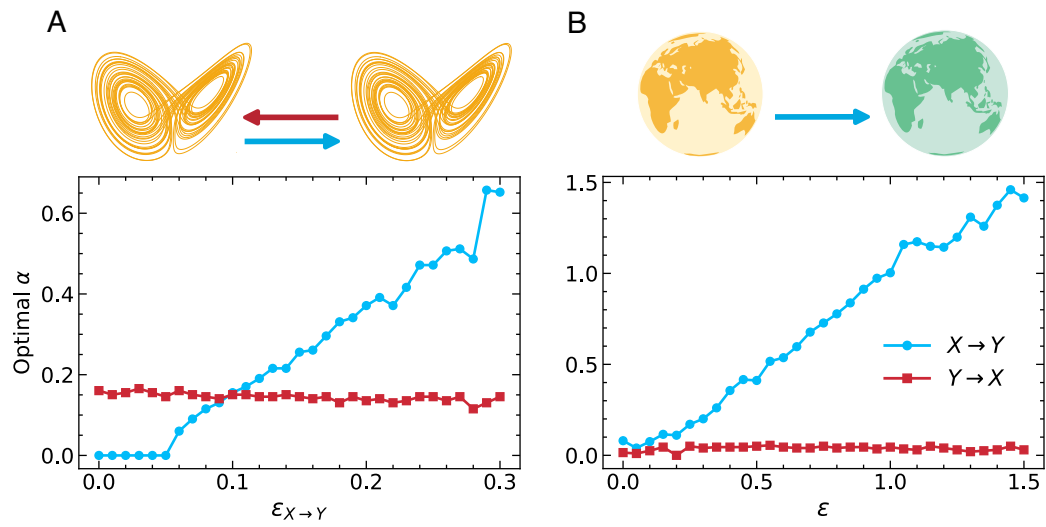


Fig. S3. Optimal α as a function of the coupling strength in direction $X \rightarrow Y$ for (A) the bidirectionally coupled Lorenz systems and (B) the unidirectionally coupled Lorenz 96 systems, using time-delay embeddings with the same parameters employed in the main text. The minimization was carried out over a grid of 300 values evenly spaced in range $[0, 1.5]$.

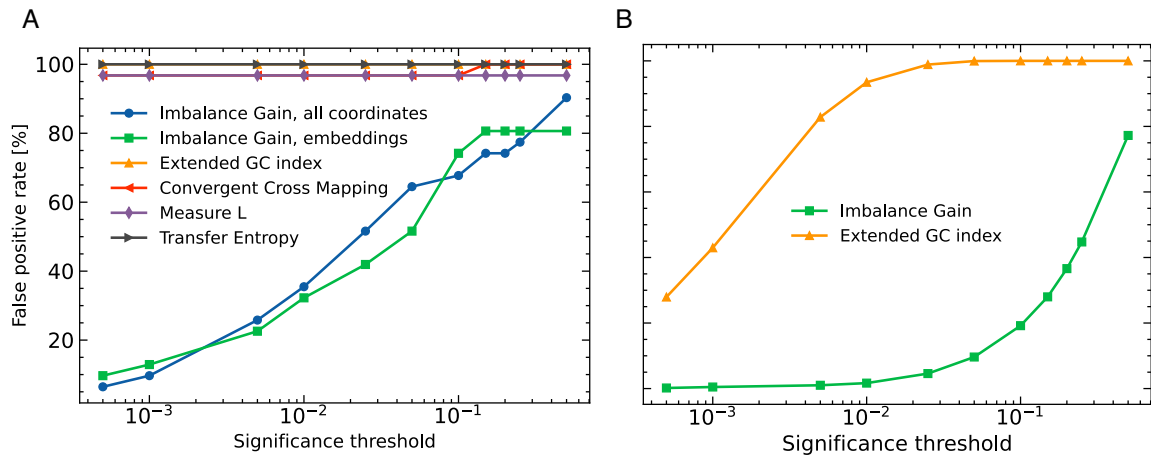


Fig. S4. False positive rates as a function of the significance threshold $\bar{p}$, in two different scenarios where causality in the ground-truth dynamics is absent: (A) in the tests $Y \rightarrow X$ for the coupled Lorenz 96 systems, and (B) in the tests after stimulus onset in the EEG experiment, referred to the link between the duration of the stimulus and the EEG activity of the 51 channels. The null hypothesis of absence of causality is rejected for $p < \bar{p}$.

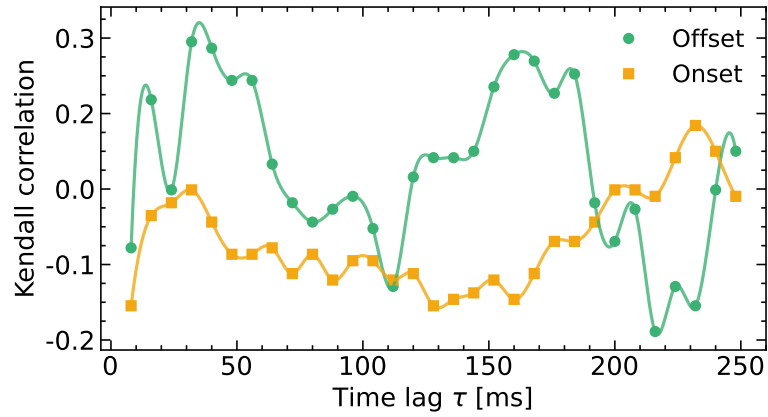


Fig. S5. Kendall correlation coefficient between the Imbalance Gains of the 19 participants in direction POz $\rightarrow$ Fz (see Fig. 4 in the main text) and their accuracies in the evaluation of the comparison stimulus' duration, as a function of the time lag τ from stimulus onset / offset. Data were interpolated with a cubic spline. The accuracy of each participant is defined as the number of correct answers divided by the total number of trials.

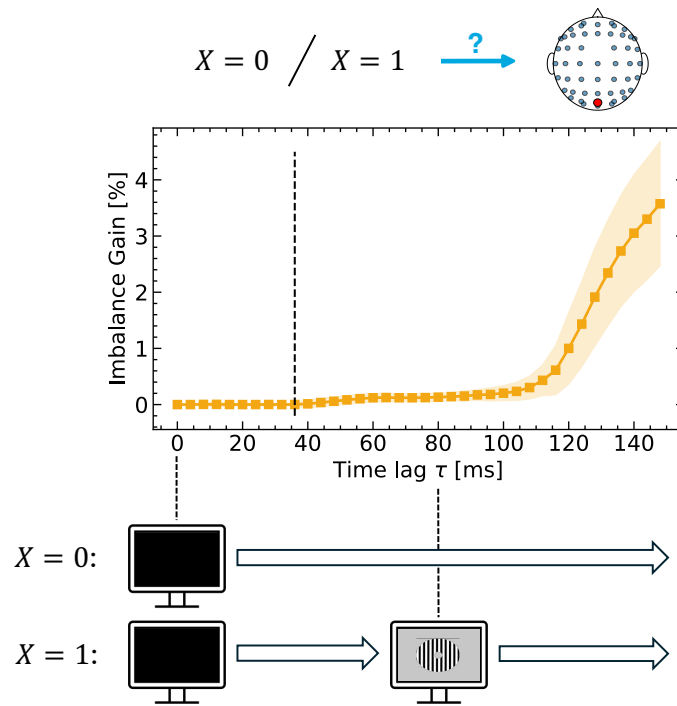


Fig. S6. Imbalance Gain as a function of the time lag τ to assess the presence of a trivial causal link between the dichotomous variable X and the EEG activity of the occipital channel POz, using time windows of 44 ms. The variable X discriminates between trials where the visual stimulus is presented after 80 ms and trials where the visual stimulus is not presented for any tested value of τ . The dashed black line corresponds to the time lag $\tau = 80 - 44$ ms, from which the predicted EEG window includes the stimulus onset for $X = 1$.

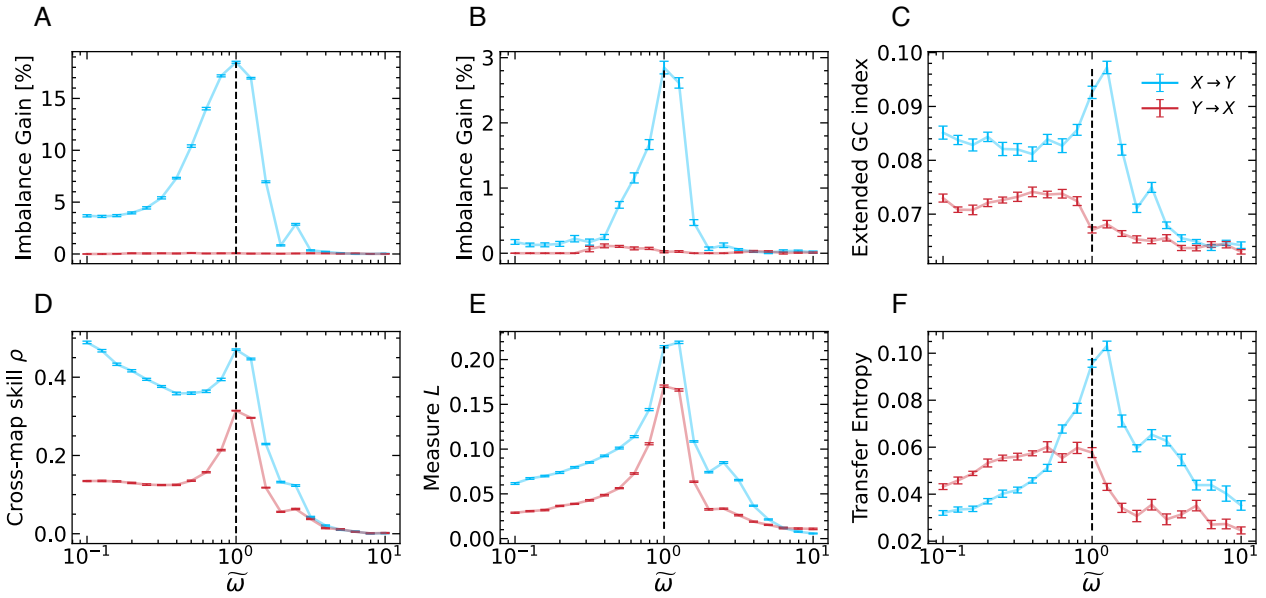


Fig. S7. Behaviors of the different causality measures against variations of the relative velocity between systems X (driver) and Y (driven), for two unidirectionally coupled Lorenz 96 systems. (A): Imbalance Gain using all the coordinates, (B): Imbalance Gain with time-delay embeddings, (C): Extended Granger Causality, (D): Convergent Cross Mapping, (E): Measure L , (F): Transfer Entropy.

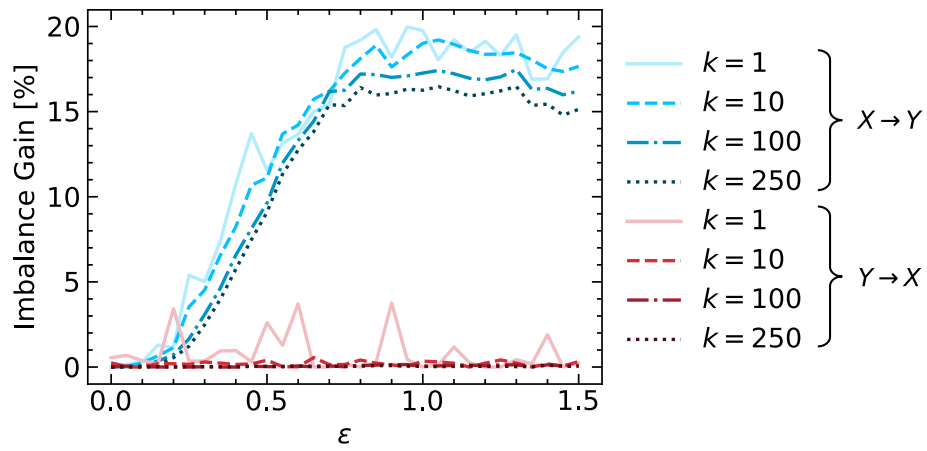


Fig. S8. Imbalance Gain as a function of the coupling strength ε for the unidirectionally coupled Lorenz 96 systems ($F_X = 5$, $F_Y = 6$), using all the coordinates of X and Y and setting $\tau = 30$.

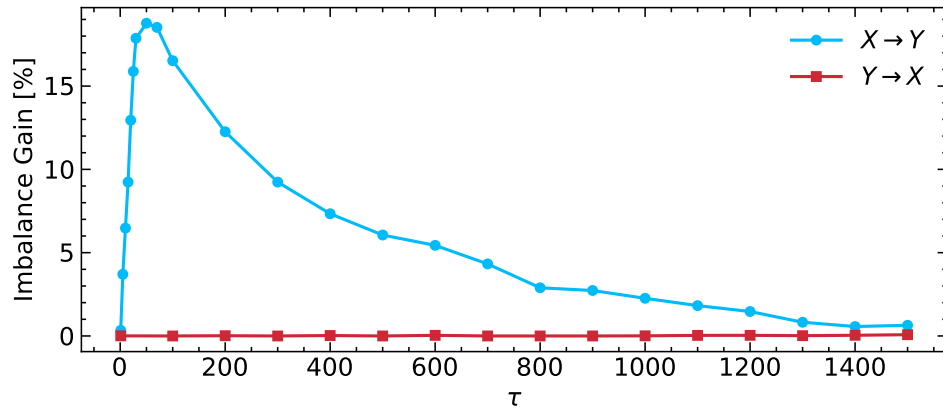


Fig. S9. Imbalance Gain as a function of the time lag τ for the unidirectionally coupled Lorenz 96 systems with $F_X = 5$, $F_Y = 6$ and $\varepsilon = 1$, using $k = 20$.

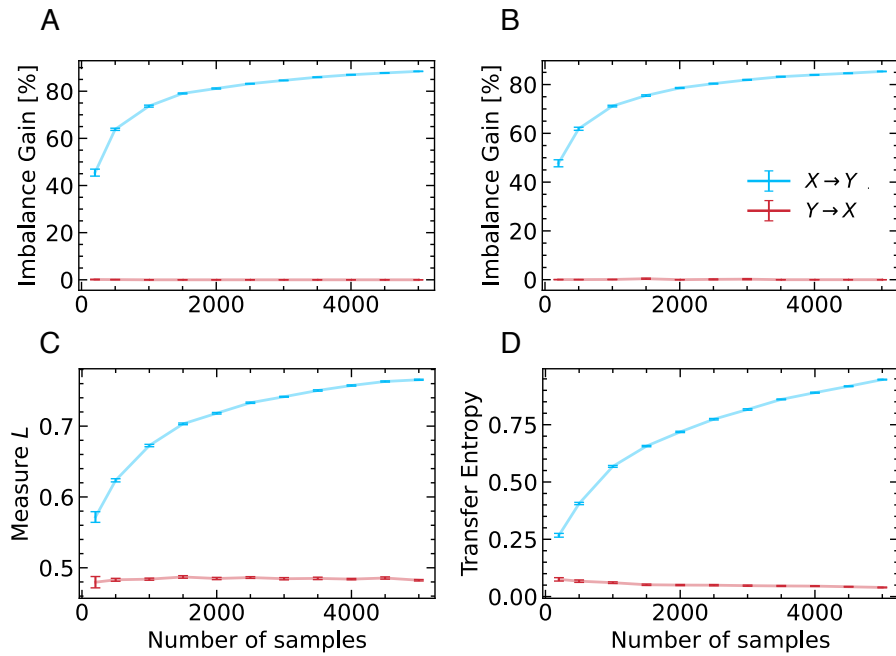


Fig. S10. Behaviors of different causality measures as a function of the number of samples N , for the different Rössler systems coupled in direction $X \rightarrow Y$. (A): Imbalance Gain using all the coordinates, (B): Imbalance Gain with time-delay embeddings, (C): Measure L , (D): Transfer Entropy. The Convergent Cross Mapping approach was excluded from the comparison as it requires to monitor the convergence as a function of the number of samples, which may not be achieved when N is small. Similarly, the Extended Granger Causality measure was not included in the comparison because the total number of points employed also depends on the number of points selected for the local regressions. Each point and its standard error was computed over 20 independent estimates.

	Coupling (detected)	No coupling (detected)
Coupling (real)	28/30 (93.3%)	2/30 (6.7%)
No coupling (real)	1/32 (3.1%)	31/32 (96.9%)

(A) Lorenz 96 systems, all coordinates

	Coupling (detected)	No coupling (detected)
Coupling (real)	25/30 (83.3%)	5/30 (16.7%)
No coupling (real)	0/32 (0%)	32/32 (100%)

(B) Lorenz 96 systems, time-delay embeddings

Table S1. Confusion matrices for the outcomes of the significance test applied to the unidirectionally coupled Lorenz 96 systems, (A) using all coordinates and (B) using time-delay embeddings. The lines of the tables report the true causal scenario (presence or absence of the coupling) while the columns refer to the outcomes of the hypothesis testing. Null distributions with 1000 samples were employed and the null hypothesis (absence of coupling) was rejected for $p < 0.001$. The class “no coupling” contains two more elements than the class “coupling” because it includes, for the particular coupling $\varepsilon = 0$, both the directions $X \rightarrow Y$ and $Y \rightarrow X$.

Movie S1. Dynamic topoplots showing the distribution of the Imbalance Gains to test the link between stimulus duration and EEG activity after stimulus offset. Each frame corresponds to a value of τ between 0 ms and 456 ms, using a stride of 12 ms.

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